

ASSIGNMENT - 4

Objective

The objective of this assignment is to understand Azure Cloud services and create an end-to-end data pipeline using Azure Storage Account and Azure Data Factory (ADF). The pipeline copies a CSV file from a source Blob container to a destination Blob container while validating file metadata.

Step 1: Create Resource Group

1. Login to Azure Portal.
2. Navigate to Resource Groups.
3. Click Create.
4. Enter Resource Group Name: **ADF-Assignment-RG**.
5. Select a region such as Central India.
6. Click Review + Create and then Create.

Purpose: Resource Group acts as a logical container for all Azure resources.

Step 2: Create Storage Account

1. Search for Storage Accounts.
2. Click Create.
3. Select the previously created Resource Group.
4. Enter Storage Account Name.
5. Choose Standard Performance and LRS Redundancy.
6. Click Review + Create.

Purpose: Storage Account stores files and data used by the pipeline.

Step 3: Create Blob Containers

Two Blob Containers are created:

Source Container

Name: **source-data**

Destination Container

Name: **destination-data**

Purpose: Source container stores input files and destination container stores processed files.

Step 4: Upload Dataset

1. Open source-data container.
2. Click Upload.
3. Select Superstore CSV file.
4. Upload the file successfully.

Purpose: This file acts as input data for the pipeline.

Step 5: Create Azure Data Factory

1. Search Azure Data Factory.
2. Click Create.
3. Select Resource Group.
4. Enter Data Factory Name.
5. Choose the same region.
6. Create the Data Factory instance.

Purpose: Azure Data Factory is used to build and manage data pipelines.

Step 6: Open Azure Data Factory Studio

After deployment:

1. Open Data Factory.
2. Click Launch Studio.
3. Explore:
 - Author
 - Monitor
 - Manage

Purpose: ADF Studio provides the development environment for creating pipelines.

Step 7: Create Linked Service

1. Go to Manage.
2. Open Linked Services.
3. Click New.
4. Select Azure Blob Storage.
5. Configure connection with Storage Account.
6. Test Connection.
7. Create Linked Service.

Purpose: Linked Service establishes connection between ADF and Azure Storage.

Step 8: Create Source Dataset

1. Go to Author.
2. Create New Dataset.
3. Select Azure Blob Storage.

4. Select Delimited Text format.
5. Choose uploaded CSV file from source-data container.
6. Import Schema.

Dataset Name: **SourceDataset**

Purpose: Represents the source file used in the pipeline.

Step 9: Create Destination Dataset

1. Create another dataset.
2. Select Azure Blob Storage.
3. Select destination-data container.

Dataset Name: **DestinationDataset**

Purpose: Represents the destination location for copied data.

Step 10: Create Pipeline

1. Click New Pipeline.
2. Name it **BlobToBlobPipeline**.

Purpose: Pipeline orchestrates the complete data movement process.

Step 11: Add Get Metadata Activity

1. Drag Get Metadata activity onto the canvas.
2. Connect it with SourceDataset.
3. Select metadata fields:
 - Exists
 - Size

- Last Modified

Purpose: Validates source file before processing.

Step 12: Add Copy Data Activity

1. Drag Copy Data activity.
2. Connect Get Metadata activity to Copy Data activity.
3. Configure:
 - Source: SourceDataset
 - Destination: DestinationDataset

Output File Name:

Superstore_Copy.csv

Purpose: Copies data from source container to destination container.

Step 13: Validate Pipeline

1. Click Validate.
2. Resolve any configuration issues.
3. Validation Successful message appears.

Purpose: Ensures pipeline configuration is correct.

Step 14: Execute Pipeline

1. Click Debug.
2. Pipeline starts execution.
3. Activities execute sequentially.

Expected Status:

Succeeded

Purpose: Tests pipeline functionality.

Step 15: Trigger Pipeline

1. Click Add Trigger.
2. Select Trigger Now.
3. Confirm execution.

Purpose: Runs the pipeline manually.

Step 16: Monitor Execution

1. Open Monitor section.
2. View execution history.
3. Check:
 - Start Time
 - End Time
 - Duration
 - Status

Expected Status:

Succeeded

Step 17: Verify Output

1. Open destination-data container.
2. Verify copied file:
Superstore_Copy.csv

Result: File is successfully copied from source container to destination container.

Result

The Azure Data Factory pipeline successfully validated the source file metadata and copied the CSV file from the source Blob container to the destination Blob container.